

Perfect Domination with Forbidden Vertices on l_k -starlike Graphs and Trees

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Abstract

Let $G(V, E)$ be a graph with n -vertex-set V and m -edge-set E . Meanwhile, each vertex v is associated with a nonnegative weight $W(v)$. Given a subset F of V , this paper studies the problem of finding a perfect dominating set D such that $\delta(D) = \sum_{v \in D} W(v)$ is minimized under the restriction that $D \subseteq (V - F)$. The vertices in F are called forbidden vertices. We first define a new class of graphs, called l_k -starlike graphs, $k \geq 1$, which is a super class of the class of starlike graphs. Then, we show that the problem is NP-Hard on l_k -starlike graphs, $k \geq 2$, but linear-time solvable when $k = 1$. Finally, an $O(n)$ -time algorithm for the problem on trees is designed.

Keywords: Domination with forbidden vertices, perfect dominating sets, l_k -starlike graphs, trees, NP-Hard

1. Introduction

Let $G(V, E, W)$ be a connected and undirected graph with n -vertex-set V and m -edge-set E , and each vertex v is associated with a nonnegative weight $W(v)$. Various domination issues on graphs have received extensive research efforts due to their theoretical importance and wide-area applications [2, 7, 8]. Given any subset Q of V , denote $\delta(Q) = \sum_{v \in Q} W(v)$.

A subset D of V is called a *dominating set* of G if there exists $u \in D$ such that $(v, u) \in E$ for each $v \in (V - D)$. Furthermore, if there exists unique $u \in D$ such that $(v, u) \in E$ for each $v \in (V - D)$, then D is called a *perfect dominating set* (PD set). This paper will deal with the following more general problem than ever.

The Restricted Weighted Minimum Perfect Dominating Set problem (the RWMPDS problem): Given a graph $G(V, E, W)$ and a subset F of V , find an PD set Q such that $\delta(Q)$ is minimized under the restriction that $Q \subseteq (V - F)$. Q is called an optimal solution of the RWMPDS problem on G .

The vertices in F are called *forbidden vertices* and F is called the *forbidden set* herein. Meanwhile, a PD set D with $D \cap F = \emptyset$ is called a *restricted perfect dominating set* (RPD set) and the problem is called the WMPDS problem if F is empty.

Fig. 1(a) depicts a graph $G_1(V_1, E_1, W_1)$ with the forbidden set $F_1 = \{x_2, x_3, x_6, x_7, x_9\}$. It is easy to check that no solution exists for the RWMPDS problem in G_1 . Fig. 1(b) shows a graph $G_2(V_2, E_2, W_2)$ with $F_2 = \{y_4, y_7, y_8\}$. It is easy to verify that $Q = \{y_1, y_5, y_6\}$ is an optimal solution with $\delta(Q) = W_2(y_1) + W_2(y_5) + W_2(y_6) = 13$.

2. l_k -starlike Graphs

An edge (u, v) of any graph G is called a *chord* of a cycle Ω of G if u and v are two nonconsecutive vertices in Ω . G is called a *chordal graph* if each cycle with length greater than three has a chord [5, 11]. Gavril demonstrated that the intersection graphs of subtrees in a tree are exactly the chordal graphs and the intersection models can be selected so that the nodes of the tree are always the *maximal cliques* of the original graphs [5]. Each vertex of the original graph then corresponds to the subtree consisting of exactly those cliques to which it belongs. The corresponding intersection model is called a *clique tree* of the graph and it can be constructed in linear-time [11]. Let T be a clique tree of the chordal $G(V, E)$ consisting of $C_1, \dots, C_\beta$ nodes. Assume that C_1 is the root of T . Denote $\text{Cliques}(v) = \{C_j \mid C_j \text{ contains } v\}$. The following lemmas can be easily established based upon the results in [11].

Lemma 1. The nodes in $\text{Cliques}(v)$ form a subtree in the clique tree T and $\sum_{j=1}^{\beta} |C_j| = \sum_{v \in V} |\text{Cliques}(v)| = O(m + n)$.

A graph is called a *split graph* if its vertices can be divided into two disjoint sets K and I such that K is a clique (complete graph) and I is an independent set [6]. Meanwhile, a chordal graph G is called *starlike* if and only of (iff) its clique tree is a star [9, 10]. We extend this definition to a more general form. Given any integer $k \geq 1$, a graph H is called a *level- k -star* (l_k -star) iff H can be obtained from a star S by inserting at most $(k - 1)$ vertices to each edge as show in Fig. 2. A chordal graph G is called *l_k -starlike* iff its clique tree is a l_k -star. By our definition, the class of starlike graphs is a subclass of the class of l_k -starlike graphs. Fig. 3 depicts the hierarchy of graphs considered in this section.

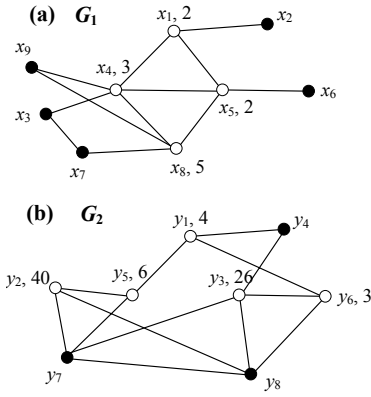


Figure 1. Two instances of the RWMPDS problem.

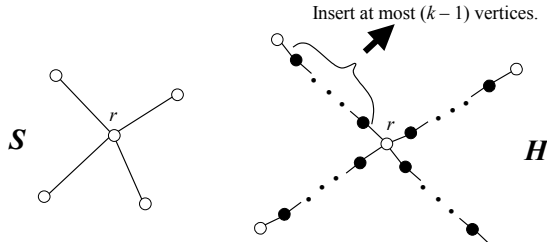
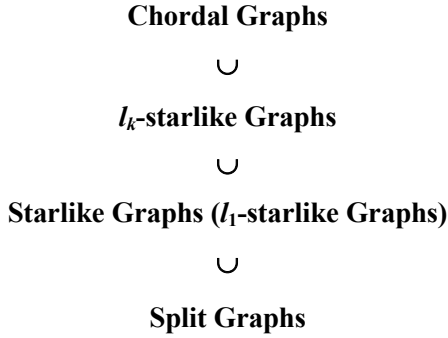

 Figure 2. Expanding a star S to a l_k -star H , $k \geq 1$.


Figure 3. The hierarchy of graphs considered in this section.

3. The RWMPDS Problem on l_k -Starlike Graphs

The WMPDS problem is NP-Hard on chordal graphs [2, 7, 8]. The WMPDS problem is a special case of the RWMPDS problem. So, the following theorem holds.

Theorem 1. The RWMPDS problem is NP-Hard on chordal graphs.

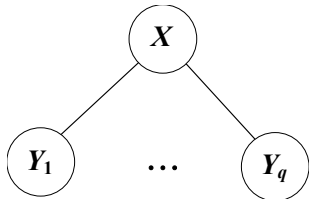


Figure 4. A clique tree of the input starlike graph.

This section will deal with the RWMPDS problem on l_k -starlike graphs. The following well-known NP-Hard problem is used for reduction [4].

The Exact Cover problem (The EC problem): Given a family of $F = \{S_1, \dots, S_p\}$ of sets where each S_i is a subset of some set $X = \{x_1, \dots, x_q\}$, does there exist a subfamily of pairwise disjoint sets of F whose union is equal to X ?

Though the following theorem may be derived using the similar technique used in [12], we show the details for the sake of the completeness and the integrity of this paper.

Theorem 2. The RMPDS problem is NP-Hard on l_k -starlike graphs, for all $k \geq 2$.

Proof: Given an instance of the EC problem in which $F = \{S_1, \dots, S_p\}$ and $X = \{x_1, \dots, x_q\}$, an input instance of the RMPDS problem $G(V, E)$ and F is constructed by the following transformation rules: (1) $V = X \cup F \cup \{a_1, \dots, a_p\}$. (2) $E = \{(x_z, S_i) \mid x_z \text{ belongs to } S_i\} \cup \{(S_j, a_j) \mid j = 1, \dots, p\} \cup \{(x_z, x_s) \mid \text{for all } z \neq s\}$. (3) $F = X = \{x_1, \dots, x_q\}$. We can easily verify that G is a l_2 -starlike graph and can be constructed in $O(q^2 + pq)$ -time.

To complete our proof, we will show that there exists a solution K of the EC problem iff there exists a RPD set Q such that $|Q| = p$. Suppose that K is a solution of the EC problem. Without a loss of generality, we can assume that $K = \{S_1, \dots, S_\alpha\}$, $1 \leq \alpha \leq p$. Then, $S_z \cap S_t = \emptyset$, for all S_z and S_t in K . It implies that each x_h in X will be dominated by one vertex in K exactly. Let $Q = K \cup \{a_{\alpha+1}, \dots, a_p\}$. Verifying that Q is a RPD set is a simple task.

Next, if Q is a RPD set with $|Q| = p$, then each x_h in X must be dominated by exactly one vertex in Q , denoted by $x_h^{(R)}$. It is easily checkable that $x_h^{(R)} \in S$, $1 \leq h \leq p$, and either $x_z^{(R)} \cap x_s^{(R)} = \emptyset$ or $x_z^{(R)} = x_s^{(R)}$, for all $z \neq s$. Therefore, $K = \cup_{1 \leq h \leq p} \{x_h^{(R)}\}$ is a solution of the EC Problem. $\square$

Corollary 1. The RWMPDS problem is NP-Hard on l_k -starlike graphs, for all $k \geq 2$.

The second result of this section is to show that the RWMPDS problem on starlike graphs can be solved in linear-time. We assume that the clique tree of input starlike graph $G(V, E)$ is as shown in Fig. 4.

Definition 1. For each Y_j , $1 \leq j \leq q$, define $\mu(Y_j)$ as follows.

$$\mu(Y_j) = \begin{cases} u_j \in (Y_j - F) \neq \emptyset, W(u_j) \leq W(v), \forall v \in (Y_j - F) \\ \text{NULL}, (Y_j - F) = \emptyset \end{cases}$$

Definition 2. For each $x \in X$, define $\Phi(x) = \text{Cliques}(x) - \{X\}$.

The following properties can be easily established.

Property 1. For any clique Ω (it may contain forbidden vertices) and any RPD set D , one of the following possibilities could occur. (1) $|D \cap X| = 0$. (2) $|D \cap X| = \{x\}$, for some $x \in X$. (3) $|D \cap X| = X$, $|X| \geq 2$.

Let D be any RPD set of G . Property 1 implies that we should handle the following three cases.

Case 1 $|D \cap X| = 0$

This case can be viewed as the case that all vertices in X are black. Consider each Y_j , for all $1 \leq j \leq q$. It is easy to see that at least one vertex in $(Y_j - (X \cup F))$ must be included in D . By the definition of perfect dominating sets and chordal graphs, it is impossible that there exists $x \in X$ such that $(\text{Cliques}(x) = \{X\})$ or $(|\text{Cliques}(x) \cap \{Y_1, \dots, Y_q\}| \geq 2)$. And we must have $(Y_j - X) - F \neq \emptyset$, for all $1 \leq j \leq q$. Since $W(v) \geq 0$, for each vertex v , it is clear that we can just only consider $D = \{\mu(Y_1), \dots, \mu(Y_q)\}$ for obtaining an optimal solution of G .

Case 2 $|D \cap X| = 1$

Assume that $(D \cap X) = \{x\}$. In this case, we can view that all vertices in $X - \{x\}$ are forbidden. We claim that $\Phi(x) = \{Y_1, \dots, Y_q\}$. All vertices in X and $(Y_j - X)$ are already dominated by x , for all $Y_j \in \Phi(x)$. If there exists $Y_i \notin \Phi(x)$, then all vertices in $(Y_i - X)$ cannot be included. But, this means that no vertex can dominate the vertices in $(Y_i - X)$. A contradiction occurs. Therefore, we must have $D = \{x\}$, for some $x \in X$ in this case.

Case 3 $|D \cap X| = X$, $|X| \geq 2$

This case implies all vertices in $(Y_j - X)$ are already dominated by some vertex in X , $1 \leq j \leq q$.

Combining the results so far, the following theorem can be obtained.

Theorem 3. The RWMPDS problem on starlike graphs can be solved in linear-time.

4. The RWMPDS Problem on Trees

This section designs an $O(n)$ -time algorithms for the RWMPDS problem on trees. Certainly, the lower bound of the RWMPDS problem is $O(n)$. Thus, our algorithm is time-optimal in worst case.

Given a tree T , we choose any vertex r to be the root and the tree will be denoted as $T(r)$. Define $\delta(T(r)) = \min\{\delta(R) \mid R \text{ is a RPD set of } T(r)\}$. Furthermore, the following definitions are made.

Also, the assumption that $W(v) \geq 0$, for all $v \in V$, implies that we just only consider $D = X$ in order to obtain an optimal solution of G .

The following algorithm can now be designed to solve the RWMPDS problem on starlike graphs correctly based upon the discussion so far.

Algorithm RWMPDS_StarlikeGraphs

Input: A starlike graph $G(V, E, W)$ and a subset F of V .

Output: A RPD set Q of G such that $\delta(Q) = \sum_{v \in D} W(v)$ is minimized.

Method:

```
{
    construct a clique tree  $T$  of  $G$ ;
    // Let its nodes be  $X, Y_1, \dots, Y_q$ .

     $Q = \text{DUMMY\_RPD\_SET}$ ;
    // We assume that  $|\text{DUMMY\_RPD\_SET}| = \infty$ .

    if not (there exists  $x \in X$  such that
         $((\text{Cliques}(x) = \{X\}) \text{ or } (|\text{Cliques}(x) \cap \{Y_1, \dots, Y_q\}| \geq 2)))$ 
        and  $((Y_j - X) - F \neq \emptyset, \text{ for all } 1 \leq j \leq q)$ 
         $Q = \{\mu(Y_1), \dots, \mu(Y_q)\}$ ;

    if ( $X$  is itself a RPD set) and  $\delta(X) < \delta(Q)$ 
         $Q = X$ ;

    for each vertex  $x \in X$  do
        if  $\Phi(x) = \{Y_1, \dots, Y_q\}$  and  $\delta(x) < \delta(Q)$ 
             $Q = \{x\}$ ;
    endfor

    return ( $Q$ );
}
```

End RWMPDS_StarlikeGraphs

Definition 3. For any non-root vertex x of $T(r)$, the following notations are defined.

- (1) Denote $T(x)$ as the subtree rooted at x and $\text{par}(x)$ is the parent of x .
- (2) $\delta_{00}(x) = \min\{\delta(R) \mid R \text{ is a RPD set of } T(x) \text{ such that } \text{par}(x) \notin R \text{ and } x \notin R\}$. The corresponding problem is denoted as the problem P_{00} .
- (3) $\delta_{01}(x) = \min\{\delta(R) \mid R \text{ is a RPD set of } T(x) \text{ such that } \text{par}(x) \notin R \text{ and } x \in R\}$. The corresponding problem is denoted as the problem P_{01} .
- (4) $\delta_{10}(x) = \min\{\delta(R) \mid R \text{ is a RPD set of } T(x) \text{ such that } \text{par}(x) \in R \text{ and } x \notin R\}$. The corresponding problem is denoted as the problem P_{10} .
- (5) $\delta_{11}(x) = \min\{\delta(R) \mid R \text{ is a RPD set of } T(x) \text{ such that } \text{par}(x) \in R \text{ and } x \in R\}$. The corresponding problem is denoted as the problem P_{11} .

Definition 4. For any vertex u of $T(r)$, the following notations are defined.

- (1) $\delta_0(T(u)) = \min\{\delta(R) \mid R \text{ is a RPD set of } T(u) \text{ such that } u \notin R\}$. The corresponding problem is denoted as the problem $P_{(0)}$.
- (2) $\delta_1(T(u)) = \min\{\delta(R) \mid R \text{ is a RPD set of } T(u) \text{ such that } u \in R\}$. The corresponding problem is denoted as the problem $P_{(1)}$.
- (3) $\delta(T(u)) = \min\{\delta(R) \mid R \text{ is a RPD set of } T(u)\}$.

Consider the root r , the following two cases will be handled, respectively.

$$\begin{aligned} & \delta_0(T(r)) \\ &= \begin{cases} \infty, Y = \emptyset \\ \sum_{1 \leq j \leq k} \delta_{00}(T(x_j)) + \min_{1 \leq j \leq q} \left\{ \delta_{01}(T(y_j)) + \sum_{1 \leq s \neq j \leq q} (\delta_{00}(T(y_s))) \right\}, Y \neq \emptyset \end{cases} \quad --(2) \end{aligned}$$

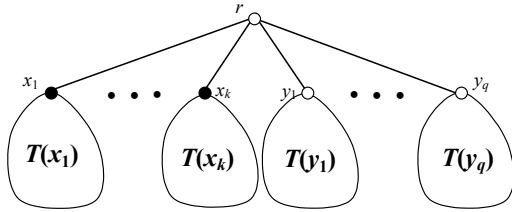


Figure 5. A tree $T(r)$, where $T(x_1), \dots, T(x_k)$ are the subtrees of r whose roots are forbidden vertices and $T(y_1), \dots, T(y_q)$ are the subtrees of r whose roots are non-forbidden vertices.

Case 2 $r \in F$, i.e., r is a forbidden vertex

Since r is a forbidden vertex, r cannot be included in any feasible solution D and if Y is not empty, then exactly one vertex in Y must be selected.

$$\begin{aligned} & \delta_{00}(T(u)) \\ &= \begin{cases} \infty, Y = \emptyset \\ \sum_{1 \leq j \leq k} \delta_{00}(T(x_j)) + \min_{1 \leq j \leq q} \left\{ \delta_{01}(T(y_j)) + \sum_{1 \leq s \neq j \leq q} (\delta_{00}(T(y_s))) \right\}, Y \neq \emptyset \end{cases} \quad --(4) \end{aligned}$$

$$\begin{aligned} & \delta_{01}(T(u)) = \delta_{11}(T(u)) \\ &= W(u) + \sum_{1 \leq j \leq k} \delta_{10}(T(x_j)) + \sum_{1 \leq j \leq q} \min\{\delta_{10}(T(y_j)), \delta_{11}(T(y_j))\} \quad --(5) \end{aligned}$$

$$\delta_{10}(T(u)) = \sum_{1 \leq j \leq k} \delta_{00}(T(x_j)) + \sum_{1 \leq j \leq q} \delta_{00}(T(y_j)) \quad --(6)$$

Case 2 $u \in V$, i.e., u is a forbidden vertex

In this case, $\delta_{01}(T(u)) = \delta_{11}(T(u)) = \infty$. The other two values, $\delta_{00}(T(u))$ and $\delta_{10}(T(u))$ can be computed by the same way as Case 1.

Case 1 $r \in (V - F)$, i.e., r is a non-forbidden vertex

By the above definitions, we can derive the following formula.

$$\delta(T(r)) = \min\{\delta_0(T(r)), \delta_1(T(r))\} \quad --(1)$$

Among all children of r , suppose that $X = \{x_1, \dots, x_k\} \subseteq F$ and $Y = \{y_1, \dots, y_q\} \subseteq (V - F)$ as shown in Figure 5. To compute $\delta_0(T(r))$, if Y is not empty, then exactly one vertex in Y must be selected. The following formula can be verified easily.

The next task is to compute $\delta_1(T(r))$. The following formula can be established.

$$\begin{aligned} & \delta_1(T(r)) = W(r) + \\ & \sum_{1 \leq j \leq k} \delta_{10}(T(x_j)) + \sum_{1 \leq j \leq q} \min\{\delta_{10}(T(y_j)), \delta_{11}(T(y_j))\} \quad --(3) \end{aligned}$$

This case is just equivalent to compute $\delta_0(T(r))$ as Case 1.

Now, consider any non-root vertex u . Without causing any confusion, among all children of u , we also use the notations $X = \{x_1, \dots, x_k\}$ and $Y = \{y_1, \dots, y_q\}$ to represent the children of u which are forbidden vertices and non-forbidden vertices, respectively. Again, the following two cases are considered, respectively.

Case 1 $u \in (V - F)$, i.e., u is not a forbidden vertex

The following formulas for the problems P_{00} , P_{01} , and P_{11} on $T(u)$ can be derived.

The problem P_{10} means that u is already dominated by $\text{par}(u)$. So, the following formula can be obtained for computing $\delta_{10}(T(u))$.

The boundary condition occurs when the subtree $T(u)$ consists of only the vertex u . Two possibilities could occur.

Case 1. u is the root of $T(r)$

The entire tree consists of only the root r , i.e., u is just the root r . The following formulas can be established.

$$\delta_0(T(u)) = \infty \text{--(7)}$$

$$\delta_1(T(u)) = \begin{cases} \infty, u \in F \\ W(u), u \notin F \end{cases} \text{--(8)}$$

$$\delta_1(T(u)) = \delta_{01}(T(u)) = \delta_{11}(T(u)) = \begin{cases} \infty, u \in F \\ W(u), u \in (V - F) \end{cases} \text{--(11)}$$

$$\delta_{10}(T(u)) = 0 \text{--(12)}$$

$$\delta(T(u)) = \min\{\delta_0(T(u)), \delta_1(T(u))\} \text{--(13)}$$

In the following, we will prove that all the above formulas can be implemented in $O(n)$ -time.

Lemma 2. Formulas (1) to (13) can be implemented in either $O(1)$ -time or $O(k + q)$ -time, except Formula (2) and Formula (4).

Lemma 3. Both Formula (2) and Formula (4) can be implemented in $O(k + q)$ -time.

Proof: $\sum_{1 \leq j \leq k} \delta_{00}(T(x_j))$ can be computed in $O(k)$ -time. Let $\delta^{(j)} = \delta_{01}(T(x_j)) + \sum_{1 \leq s \neq j \leq q} \delta(T(y_s))$, $1 \leq j \leq q$. It is easy to verify that $\delta^{(j+1)} = \delta^{(j)} - \delta_{01}(T(x_j)) + \delta_{01}(T(x_{j+1})) - \delta(T(x_{j+1})) + \delta(T(x_j))$, $2 \leq j \leq q$. Therefore, $\delta^{(1)}, \dots, \delta^{(q)}$ can be computed in $O(q)$ -time. This implies that both Formula (2) and Formula (4) can be implemented in $O(k + q)$ -time. $\square$

Finally, an optimal solution can be identified by examining each vertex once from the root r after $\delta(T(r))$ has been obtained and its time-complexity is clear $O(n)$.

Theorem 4. The RWMPDS problem can be solved in $O(n)$ -time on trees.

5. Conclusions

This paper first defined the class of l_k -starlike graphs, $k \geq 1$, which is a super class of starlike graphs. Then, we shown that the RWMPDS problem is NP-Hard on l_k -starlike graphs, $k \geq 2$, but linear-time solvable when $k = 1$. Finally, an $O(n)$ -time algorithm for the problem on trees was also designed.

In the future, it is worthy to identify the classes of graphs such that the time-complexities of the domination with forbidden sets and no forbidden sets fall into different complexity classes. Studying other types of domination problems, such as independent domination, total domination, and connected domination with forbidden vertices, is another

$$\delta(T(u)) = \min\{\delta_0(T(u)), \delta_1(T(u))\} \text{--(9)}$$

Case 2. u is non-root

The following formulas can be established.

$$\delta_0(T(u)) = \delta_{00}(T(u)) = \infty \text{--(10)}$$

research direction that deserves us to study in detail in the future.

Acknowledgements

This research was supported by National Science Council, Taiwan, R.O.C., under contract number NSC-94-2213-E-128-005.

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